

# Detection of Fake News & Trustworthiness In Social Media Using Machine Learning via Textual Stylometry

\*Term Project Research Paper

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**Abstract**—The rapid proliferation of digital misinformation presents a significant challenge to the integrity of democratic processes and social media ecosystems. This study develops a robust, content-agnostic machine learning system for automated fake news detection, prioritizing Textual Stylometry over volatile semantic claims. By focusing on structural "fingerprints" of deceptive writing, our model remains effective across diverse topics. The study implements a comprehensive feature engineering pipeline consisting of nine distinct attributes: word count, sentiment polarity, subjectivity, lexical diversity, capitalization ratio, exclamation frequency, average word length, question mark count, and digit ratio. Through strategic pruning and hyperparameter optimization, the generalization gap was reduced from 16.91% to 4.80%, achieving a peak accuracy of 82.16% and an AUC of 0.90.

**Index Terms**—Fake News Detection, Machine Learning, Stylometry, Neural Networks, Overfitting Mitigation, Data Augmentation.

## I. INTRODUCTION

In the modern digital information landscape, social media platforms have emerged as primary channels for news consumption. However, the absence of traditional editorial oversight has facilitated the viral spread of "fake news"—content intentionally designed to mislead. The volume and velocity of this misinformation surpass human fact-checking capabilities, necessitating automated, scalable computational solutions.

Traditional approaches often rely on Natural Language Processing (NLP) to analyze semantic claims or verify facts against external databases. While effective, these methods are often topic-dependent and struggle with newly emerging subjects where ground-truth data is unavailable. This research addresses the problem through the lens of Textual Stylometry. Our core hypothesis is that deception leaves a unique linguistic signature—characterized by emotional intensity, simplified vocabulary, and specific punctuation usage—that remains constant regardless of the topic. The primary objective is to build a content-agnostic classification system that determines trustworthiness based on writing style rather than semantic claims.

## II. RELATED WORK

The following 15 academic works provide the theoretical foundation for this research:

1) **Vogel and Jiang (2019)**: This study introduces the concept of "Linguistic Fingerprinting" for detecting deceptive content. The authors argue that misinformation authors leave a unique stylistic signature that transcends the news topic itself. By focusing on fine-grained features such as auxiliary verb frequency and sentence complexity, they proved that stylistic cues are powerful discriminators in content-agnostic classification.

2) **Kozłowski et al. (2020)**: Focusing on "Readability Indices," this study proved that fake news is often written at a lower readability level to ensure quick consumption and viral spread. The researchers utilized metrics like the Gunning Fog Index to quantify structural simplicity. Their findings justify the use of word count and average word length as primary indicators of deceptive reporting.

3) **Amado et al. (2018)**: This paper introduces a stylometric framework specifically designed for detecting political misinformation. It analyzes "Functional Words" (prepositions and conjunctions) as stable markers that deceptive writers cannot easily manipulate. Their success in cross-domain detection supports the goal of creating a system that is not tied to specific keywords.

4) **Gravanis et al. (2019)**: In this benchmarking study, the authors investigated the "Cognitive Load" of writing fake news. They hypothesized that the act of deception requires more cognitive effort, which manifests as changes in linguistic diversity and sentence structure. Their analysis found that fake news often has a higher ratio of adverbs and shorter sentences compared to factual news.

5) **Türkmen and Özcan (2021)**: This research analyzed the impact of morphological and stylistic features in the Turkish news landscape. The authors found that exclamation frequency and capitalization ratios are highly significant in Turkish

clickbait and fake news. This study highlights the universal nature of stylometric patterns across different languages and regions.

6) **Zhou et al. (2020)**: This study explores the "Explainability" of stylometric detectors using LIME and SHAP frameworks. The authors emphasize that understanding why a model flags an article—such as high digit ratios or extreme sentiment—is critical for user trust. This research motivated the transition toward more transparent and interpretable AI models.

7) **Bondielli and Marcelloni (2019)**: Focusing on the "Temporal Evolution" of fake news writing styles, this study suggests that misinformation tactics evolve over time to bypass filters. They recommend using ensemble architectures like Gradient Boosting to capture these changing patterns. This influenced our decision to benchmark multiple diverse machine learning models.

8) **Kaliyar et al. (2021)**: This research highlights the issue of "Over-Parameterization" in deep learning models. The authors argue that overly complex models tend to memorize noise in training data, leading to a failure in real-world generalization. This served as a critical guide for our strategy to use model pruning to control complexity.

9) **Giachanou et al. (2019)**: The authors investigated the relationship between "Emotional Intensity" and news credibility. They developed a model incorporating emotional signals like empathy and fear into stylometric analysis. Their finding that subjectivity and sentiment polarity are the most stable features for detection is a cornerstone of our methodology.

10) **Zhang and Ghorbani (2020)**: This study provides a comprehensive overview of "Adversarial Stylometry," where deceptive writers attempt to mask their style. The researchers suggested that digit ratios and average word complexity are the hardest features for authors to artificially manipulate. This discovery justifies our focus on these specific advanced features.

11) **Rubin and Lukoianova (2018)**: Differentiating between satire and malicious misinformation, this study introduces the concept of "Factual Density." They found that real news has a higher density of numerical data and dates compared to satirical content. This justifies our use of the digit ratio feature to distinguish between satire and fake news.

12) **Shu et al. (2019)**: Investigating the "Social Context" of news, this research explores how metadata can complement stylometry. Their analysis of source bias through scraping diverse domains influenced our data collection strategy. By using a heterogeneous corpus, we aimed to prevent domain-bias and improve overall model performance.

13) **Hanselowski et al. (2018)**: In this large-scale study, the authors benchmarked several architectures, emphasizing that the "Structural Layout" of an article provides crucial evidence of authenticity. They found that capitalization ratios in body text are highly predictive. This supports our inclusion of capitalization frequencies as a core stylometric feature.

14) **Thorne et al. (2019)**: This study discusses the "Verification Gap" between automated systems and human fact-

checkers. They argue that stylometry should be used as a "Pre-Filter" to flag high-risk content for human review. This vision aligns with our project's goal of creating a fast and efficient filtering system for social media.

15) **Biyani et al. (2016)**: As one of the earliest works focusing specifically on "Clickbait Stylometry," the authors proved that the presence of interrogative sentences and sensationalist punctuation are the primary drivers of low-credibility clicks. Their findings confirm that provocative punctuation is a vital indicator in the modern misinformation landscape.

### III. DATA AND FEATURE ANALYSIS

#### A. Dataset Integration and Augmentation

The integrity of the machine learning system relies on a diverse and balanced dataset. We merged four distinct news corpora, including *FakeNewsNet* and *WELFake*, to establish a comprehensive foundation for training. A significant challenge was the initial class imbalance and the lack of representative satirical content. To address this, we implemented targeted data augmentation using the *newspaper3k* library. We scraped 152 additional verified satirical and misinformation articles from domains like *The Onion* and *The Borowitz Report*. This ensured the system could distinguish between "humorous satire" and "malicious fake news," which often share stylistic similarities but differ in factual density.

#### B. Class Categorization and Labeling

The project utilizes a binary classification scheme to categorize articles based on their trustworthiness. Understanding these classes is fundamental to interpreting the model's performance:

**1) Class 0: Reliable (Negative) News:** This class represents verified, objective, and high-trustworthiness content. Linguistically, these articles exhibit lower subjectivity, neutral sentiment polarity, and professional syntax. They are characterized by a high "Digit Ratio" due to the frequent citation of dates, statistics, and verifiable data points. Sources typically include established news agencies such as Reuters and the Associated Press.

**2) Class 1: Fake/Deceptive (Positive) News:** This class encompasses misinformation, propaganda, and sensationalist clickbait. These articles often display high emotional intensity (high sentiment polarity) and extreme subjectivity. They frequently utilize "shouting" techniques, such as high capitalization ratios and excessive exclamation marks, to trigger emotional responses. This class also includes satirical articles from sources like *The Onion*; while satire is not intended to deceive in the same way as propaganda, it shares the hyperbolic and non-neutral stylistic signatures that the model is trained to flag as "non-traditional" reporting.

#### C. In-Depth Stylometric Feature Engineering

A specialized pipeline was developed to extract nine stylometric features. These features were chosen based on the psychological and linguistic markers of deception:

- **Sentiment Polarity:** Measures emotional orientation from -1 (negative) to 1 (positive). Fake news often employs extreme polarity to provoke anger or excitement.
- **Subjectivity:** Quantifies personal opinion vs. facts (0 to 1). High scores indicate biased interpretations rather than objective facts.
- **Lexical Diversity:** The Type-Token Ratio. Deceptive writing often suffers from limited vocabulary and repetitive phrasing.
- **Word Count:** Total length. Sensationalist news is often shorter to capture short attention spans, whereas authentic reports are more comprehensive.
- **Average Word Length:** A proxy for complexity. Professional news employs more formal terminology compared to simpler fake news language.
- **Exclamation Frequency:** A marker of "clickbait" and sensationalism used to create an artificial sense of urgency.
- **Question Mark Count:** Indicates rhetorical questioning or speculative headlines used to imply claims without evidence.
- **Digit Ratio:** Real news is denser in digits due to numerical data citation, which misinformation often lacks.
- **Capitalization Ratio:** High ratios are indicative of sensationalist "shouting" in body text.

#### D. Statistical Validation

To prove these features are not random, we performed Independent Two-Sample T-tests. Every feature yielded a p-value less than 0.001, confirming high statistical significance and justifying their use in modeling.

### IV. METHODOLOGY

The technical framework follows a modular pipeline designed for high robustness. We benchmarked nine architectures, ranging from linear models to complex ensemble structures.

#### A. Overfitting Mitigation through Pruning

Initial testing identified a generalization gap of 16.91% in Random Forest. We implemented a "Pruning" strategy, limiting the maximum depth to 10 and increasing minimum samples per leaf. This forced the model to learn broader linguistic patterns, successfully reducing the gap to 4.80%.

#### B. High-Performance Artificial Neural Network (ANN)

The most successful architecture was the Artificial Neural Network, specifically the Multilayer Perceptron (MLP). It successfully captured non-linear relationships between the nine stylometric features.

**1) Architecture and Regularization:** The ANN utilized multiple hidden layers to process the vector. Regularization was critical in reducing the initial 16.91% generalization gap down to a remarkable 1.31% for this specific model.

**2) Training and Optimization:** We used *GridSearchCV* to fine-tune hyperparameters. The training involved 10-fold cross-validation to ensure the stability of the weights.

**3) Performance:** The final ANN achieved a test accuracy of 82.16% and an AUC of 0.90. The model correctly identified 1,736 real articles and 1,617 fake ones, maintaining a balanced F1-score of 0.82-0.83.

### V. EXPERIMENTAL RESULTS

The evaluation phase results are detailed in the tables below.

TABLE I  
BASELINE MODEL PERFORMANCE (INITIAL PHASE)

| Model               | Mean CV Accuracy |
|---------------------|------------------|
| LOGISTIC REGRESSION | 72.51%           |
| K-NEAREST NEIGHBORS | 73.78%           |
| RANDOM FOREST       | 79.05%           |

TABLE II  
OPTIMIZED MODEL PERFORMANCE (10-FOLD CV)

| Model               | Mean CV Accuracy |
|---------------------|------------------|
| LOGISTIC REGRESSION | 74.82%           |
| K-NEAREST NEIGHBORS | 73.78%           |
| RANDOM FOREST       | 79.21%           |

TABLE III  
STATISTICS REPORT OF OPTIMIZED RANDOM FOREST

| Class           | Precision | Recall | F1-Score    | Support     |
|-----------------|-----------|--------|-------------|-------------|
| 0 (Negative)    | 0.80      | 0.79   | 0.80        | 2091        |
| 1 (Positive)    | 0.78      | 0.79   | 0.79        | 1990        |
| <b>Accuracy</b> |           |        | <b>0.79</b> | <b>4081</b> |

TABLE IV  
CV ACCURACIES AFTER NEW FEATURE ENGINEERING

| Model                     | Mean CV Accuracy | Std. Dev. |
|---------------------------|------------------|-----------|
| Logistic Regression (LR)  | 77.82%           | 0.031     |
| K-Nearest Neighbors (KNN) | 77.82%           | 0.042     |
| Random Forest (RF)        | 82.61%           | 0.019     |

### VI. DISCUSSION

The results validate the effectiveness of Textual Stylometry. The successful reduction of the generalization gap proves that our strategies forced models to learn fundamental writing styles. Sentiment polarity and punctuation frequency emerged as the most influential discriminators, confirming that misinformation relies heavily on emotional provocation.

### VII. CONCLUSION

This project developed an automated system for identifying misinformation with an accuracy exceeding 82%. The methodology effectively addressed overfitting and statistically validated all features. Future directions involve integrating this system into real-time social media filters.

## REFERENCES

TABLE V  
STATISTICS REPORT OF RANDOM FOREST (NEW FEATURES)

| Class           | Precision | Recall | F1-Score    | Support     |
|-----------------|-----------|--------|-------------|-------------|
| 0 (Negative)    | 0.81      | 0.84   | 0.83        | 2091        |
| 1 (Positive)    | 0.82      | 0.80   | 0.81        | 1990        |
| <b>Accuracy</b> |           |        | <b>0.82</b> | <b>4081</b> |

TABLE VI  
COMPARISON BEFORE OVERFIT MITIGATION

| Model                | Accuracy |
|----------------------|----------|
| Neural Network (ANN) | 0.8216   |
| Random Forest        | 0.8189   |
| Gradient Boosting    | 0.8149   |
| Decision Tree        | 0.7833   |
| AdaBoost             | 0.7804   |
| SVM                  | 0.7797   |
| Logistic Regression  | 0.7743   |
| Naive Bayes          | 0.6133   |

TABLE VII  
COMPARISON AFTER OVERFIT MITIGATION

| Model                | Test Set Accuracy |
|----------------------|-------------------|
| Neural Network (ANN) | 82.16%            |
| Gradient Boosting    | 81.50%            |
| Random Forest        | 81.35%            |
| AdaBoost             | 78.04%            |
| SVM                  | 77.97%            |
| KNN                  | 77.51%            |
| Logistic Regression  | 77.43%            |
| Decision Tree        | 76.72%            |
| Naive Bayes          | 61.33%            |

TABLE VIII  
STATISTICAL REPORT OF NEURAL NETWORKS MODEL (ANN)

| Class           | Precision | Recall | F1-Score    | Support     |
|-----------------|-----------|--------|-------------|-------------|
| 0 (Negative)    | 0.82      | 0.83   | 0.83        | 2091        |
| 1 (Positive)    | 0.72      | 0.81   | 0.82        | 1990        |
| <b>Accuracy</b> |           |        | <b>0.82</b> | <b>4081</b> |

TABLE IX  
FINAL OVERFIT MITIGATION AND GENERALIZATION GAP

| Model           | Train Acc. | Test Acc. | Gap   | Status  |
|-----------------|------------|-----------|-------|---------|
| Logistic Reg.   | 77.95%     | 77.43%    | 0.52% | Healthy |
| KNN             | 81.06%     | 77.51%    | 3.53% | Healthy |
| Random Forest   | 86.15%     | 81.35%    | 4.80% | Healthy |
| SVM             | 78.82%     | 77.97%    | 0.85% | Healthy |
| Neural Network  | 83.47%     | 82.16%    | 1.31% | Healthy |
| Gradient Boost. | 83.18%     | 81.50%    | 1.68% | Healthy |
| AdaBoost        | 79.03%     | 78.04%    | 0.99% | Healthy |
| Decision Tree   | 78.32%     | 76.72%    | 1.60% | Healthy |
| Naive Bayes     | 62.07%     | 61.33%    | 0.74% | Healthy |

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